INCH-POUND

MIL-T-50335B (AR)
30 January 1992
SUPERSEDING
MIL-T-50335A (AR)
28 November 1984

MILITARY SPECIFICATION

TEST SET, POWER SUPPLY: MX-8051/TPM-22A

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements and tests for the Power Supply and Servo Test Panel, Subassembly MX-8051, which are part of Radar Test Set AN/TPM-22A (Support Maintenance Shop Set). This panel contains equipment for testing the power supplies and servo electronics of the Radar Set AN/VPS-2 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the Issues of these documents are those listed in the Issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-F-13926 - Fire Control Materiel, Manufacture and Inspection, General Specification for
MIL-I-45607 - Inspection Equipment, Acquisition, Assemblies
MIL-C-50330 - Converter DC/DC: 10549021

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines
MIL-STD-2000 - Standard Requirements for Soldering Electrical and Electronic Assemblies

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (see 6.4)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

10549021 - DC-DC Converter
10554530 - Power Supply and Servo Test Panel
10554539 - Wire Runs List
10556303 - W96 Cable Assembly

PACKAGING DATA SHEET

PS 10554530 - Packaging of Power Supply and Servo Test Panel

(Copies of drawings, publications, and other Government documents required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

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3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 10554530, all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the Power Supply and Test Panel shall meet the requirements of MIL-F-13926 as follows:

a. Dimensions and tolerances
b. Inorganic protective surface finishings
c. Part identification and marking
d. Workmanship

The following subassembly shall have been inspected and accepted in accordance with the specification prior to assembly in the power supply and servo test panel.

UNIT NO.	NAME	DRAWING	SPECIFICATION
A1	DC/DC Converter Assembly	10549021	MIL-C-50330

3.3 Environmental service conditions.

3.3.1 vibration. The Power Supply and Test Panel shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to sinusoidal vibrations, in each of three mutually perpendicular axes, as follows:

Frequency Range	Amplitude
5 to 15 hertz	0.06 inches double amplitude
15 to 25 hertz	0.04 inches double amplitude
25 to 55 hertz	0.02 inches double amplitude

The frequency of vibration shall be swept logarithmically.
The cycling time shall be 120 minutes per axis.
The sweep rate shall be 3.75 minutes ± 10% per decade.
The 5-55-5 hertz sweep time is 7 minutes 35 seconds ± 10%.

3.3.2 Shock. The Power Supply and Test Panel shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to a total of 18 impulse shocks consisting of three shock impulses of 15 g's ± 10%, 11 milliseconds ± 10%, half-sine wave applied in each direction of 3 mutually perpendicular axes.

3.3.3 Altitude.

3.3.3.1 Operating. The Power Supply and Test Panel shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches of Hg).

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3.3.3.2 Non-operating. The Power Supply and Test Panel shall meet the requirements of 3.4 and 3.5 after exposure for not less than 10 minutes to barometric pressure encountered at 40,000 feet above sea level (5.5 inches of Hg).

3.3.4 Temperature.

3.3.4.1 Operating. The Power Supply and Test Panel shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between +32°F and +131°F.

3.3.4.2 Non-operating (storage). The Power Supply and Test Panel shall not suffer permanent damage when subjected to the constant temperature extremes of -60°F for a period of 24 hours and +160°F for a period of 4 hours (after temperature stabilization), and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (non-operating). The Power Supply and Test Panel shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for period of 10 days.

3.4 Electrical.

3.4.1 Continuity. Each conductor shall be electrically continuous as specified on the wire runs list (10554539).

3.4.2 Resistance.

5.4.2.1 Overload test circuitry. Resistance values, control settings and connection points shall be as indicated in Table I.

3.4.2.2 Converter loading circuitry. Resistance between VDC terminals (TP27 and TP28) of the power supply section shall be as indicated in Table II with TEST SELECTOR switch S13 In CONVERTER position and under conditions given in Table II.

3.4.2.3 Servo electronics loading circuitry. Resistance values between points shall be as indicated in Table III.

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3.5 Performance.

3.5.1 Power supply test section.

3.5.1.1 +24 VDC switch S1.

3.5.1.1.1 OFF legend. The OFF legend of S1 shall be illuminated when 24 Vdc + 5% is applied between J10-A(+) and J10-I(Ret) and S1 is in normally closed (OUT) position.

3.5.1.1.2 ON legend. The ON legend of S1 shall be illuminated when 24 Vdc ± 5% is applied between J10-A(+) and J10-I(Ret) and S1 is in normally open (IN) position.

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TABLE I. Overload test circuitry resistance

TERMINAL POINTS	LOAD SWITCH 0-15A LOAD (S10)	LOAD SWITCH 0-2A LOAD (S11)	VARIABLE LOADS 0-15A LOAD (R27)	VARIABLE LOADS 0-2A LOAD (R29)	RESISTANCE BETWEEN TERMINAL POINTS OHMS ±10%
0-15A LOAD (E17. E18)	MIN		Extreme CW		30
0-15A LOAD (E17, E18)	MAX		Extreme CW		less than 1
0-15A LOAD (E17, E18)	MAX		Extreme CW		3
0-15A LOAD (E17, E18)	NL				greater than 10 meg
0-2A LOAD (E19, E20)		MIN		Extreme CW	500
0-2A LOAD (E19, E20)		MAX		Extreme CW	less than 1
0-2A LOAD (E19, E20)		MAX		Extreme CW	50
0-2A LOAD (E19, E20)		NL			greater than 10 meg

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TABLE II. Converter loading circuitry resistance.

CONVERTER SWITCH S14 POSITION	LOAD SWITCH SETTING	RESISTANCE BETWEEN VDC TERMINALS (TP27, TP28) OHMS ± 5%
OFF		Less than 0.5
+24 V		18*
+22 V	±22 V UNREG LOAD: NL	greater than 10 meg
+22 V	±22 V UNREG LOAD: FL	22
-22 V	±22 V UNREG LOAD: NL	greater than 10 meg
-22 V	±22 V UNREG LOAD: FL	22
+30 V	±30 V UNREG LOAD: NL	greater than 10 meg
+30 V	±30 V UNREG LOAD: FL	30
-30 V	±30 V UNREG LOAD: NL	greater than 10 meg
-30 V	±30 V UNREG LOAD: FL	30
+40 V	±40 V UNREG LOAD: NL	greater than 10 meg
+40 V	±40 V UNREG LOAD: FL	40
-40 V	±40 V UNREG LOAD: NL	greater than 10 meg
-40 V	±40 V UNREG LOAD: FL	40
+40 V	+40 V, +14 V UNREG LOAD: NL	greater than 10 meg
+40 V	+40 V, +14 V UNREG LOAD: FL	40
+24 V	+40 V, +14 V UNREG LOAD: NL	greater than 10 meg
+24 V	+40 V, +14 V UNREG LOAD: FL	greater than 10 meg

*Tolerance is ± 12 ohms.

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TABLE III. Servo electronics loading circuitry resistance.

TERMINAL POINTS					RESISTANCE BETWEEN POINTS
J1-m	SERVO AMPL	OUTPUT	H	(TP17)	1000 ohms ± 1%
J1-r	SERVO AMPL	OUTPUT	L	(TP18)	1000 ohm ± 1%
J7-a	ELEVATION	OUTPUT	H	(TP19)	1000 ohms ± 1%
J7-m	ELEVATION	OUTPUT	L	(TP20)	1000 ohms ± 1%
J7-g	TRAVERSE	OUTPUT	H	(TP21)	1000 ohms ± 1%
J7-g	TRAVERSE	OUTPUT	L	(TP22)	1000 ohms ± 1%
J1-m	J1-r				5 ohms ± 5%
J7-a	J7-m				5 ohms ± 5%
J7-g	J7-g				5 ohms ± 5%

3.5.1.2 TEST MODE switch S4.

3.5.1.2.1 POWER SUPPLY CONV & SERVO legend. The POWER SUPPLY CONV & SERVO legend of S4 shall be illuminated and 24 Vdc +5% shall be present between J5-A(+) and J5-B(Ret) under the following conditions:

a. Conditions of 3.5.1.1.2 are met.

b. 24 Vdc ± 5% applied between J10-B(+) and J10-I(Ret).

c* DC circuit breaker CB1 is closed (white band not visible).

d. S4 is in normally open (IN) position.

3.5.1.2.2 REGULATOR legend. The REGULATOR legend of S4 shall be illuminated and the output voltages listed in Table IV shall be present at VOLTS meter (M2) for each CONVERTER rotary switch and load toggle switch position specified in Table IV under the following conditions:

a. W-96 Cable (Item 6) is connected between J2 and J3.

b. conditions of 3.5.1.1.2 are met.

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c. 24 vdc ± 5% is applied between J10-B(+) and J10-I(Ret).

d. DC circuit breaker CB 1 is closed (white band not visible).

e. S4 is in normally open (IN) position.

TABLE IV. VOLTS meter readings.

CONVERTER SWITCH	LOAD SWITCH SETTING	VOLTS METER INDICATION (VOLTS)
+22 V, -22 V	+22 V UNREG LOAD: FL	19.8 to 24.2
+30 V, -30 V	±30 V UNREG LOAD: FL	27.0 to 33.0
+40 V, -40 V	±40 V UNREG LOAD: FL	36.0 to 44.0
+40 V	+40 V, +14 V UNREG LOAD: FL	36.0 to 44.0
+24 V	+40 V, +14 V UNREG LOAD: FL	20.0 to 26.4

3.5.103 DC circuit breaker CB1. The DC circuit breaker (CB1) shall be closed when reset button is depressed so white band is not exposed. The circuit breaker shall be open when reset button is pulled out to expose white band.

3.5.1.4 INTERLOCK switch S2.

3.5.1.4.1 STANDBY legend. The STANDBY legend of S2 shall be illuminated when 24 Vdc ± 5% is applied between J9-G(+) and J9-H (Ret).

3.5.1.4.2 OPERATE legend. The OPERATE legend of S2 shall be illuminated when 24 Vdc ± 5% is applied between J9-V(+) and J9-a (Ret). Continuity shall exist between J9-F and J9-H when S2 is in normally open (IN) position. Resistance between J9-F and J9-H shall be infinite when S2 is in normally closed (OUT) position.

3.5.1.5 RADIATE COMMAND switch S3.

3.5.1.5.1 RADIATE COMMAND legend. The RADIATE COMMAND legend of S3 shall be illuminated when 24 Vdc ± 5% is applied to J8-N(+) and J8-P(Ret).

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3.5.1.5.2 RADIATE COMMAND switch. With S3 in normally open (IN) position and conditions are as stated In 3.5.1.2.1, 24 Vdc + 5% shall be present between J6-U(+) and J6-V(Ret). With S3 in normally closed (OUT) position, 0.0 ± 0.4 Vdc shall be present between J6-U(+) and J6-V(Ret).

3.5.1.6 Regulator loading and monitoring. AMPERES (M1) and VOLTS (M2) meters shall indicate as shown in Table V under the following conditions:

a. TEST SELECTOR Switch S13 in POWER SUPPLY position.

b. Controls and conditions met in accordance with 3.5.1.2.1.

c. A resistive load of 2.9 ohms ±10% between J5-A and J5-B.

d. Voltages applied to J8 and J9 as follows:

J8	VOLTAGE
X	+15 V ± 3%
FF	15 V Ret
Y	-15 V ± 3%
GG	+23 V ± 3%
CC	23 V Ret
RR	-23 V ± 3%
VV	+33 V ± 3%
MM	+33 V Ret
CC	-33 V ± 3%
UU	-33 V Ret
A and B	+6 V ± 5%
C and V	6 V Ret

J9	VOLTAGE
X and V	+29 V ± 1%
Y and W	29 V Ret

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TABLE V. AMPERES and VOLTS meter readings

POWER SUPPLY SWITCH S12 POSITION	LOAD SWITCH SETTING		AMPERES METER (M1)	VOLTS METER (M2)**	TERMINAL POINTS
OFF			0	0	
+24 V 0-50 A			6.9 - 10.0	24 ± 7%	
+15 V 0-2 A	± 15 V REG LOAD:	FL	1.0 ± 8%	15 ± 9%	+15 V Red (+).
		NL	0		+15 V Black (-)
		CLL	1.2 ± 7%		
-15 V 0-2 A	± 15 V REG LOAD:	FL	1.0 ± 8%	15 ± 9%	-15 V Red (-)
		NL	0		-15 V Black (Ret)
		CLL	1.2 ± 7%		
+23 V 0-2 A	± 23 V REG LOAD:	FL	1.2 ± 7%	23 ± 7%	+23 V Red (+)
		NL	0		+23 V Black (Ret)
		CLL	1.36 ± 7%		
-23 V 0-2 A	± 23 V REG LOAD:	FL	1.2 ± 7%	23 ± 7%	-23 V Red (-)
		NL	0		-23 V Black (Ret)
		CCL	1.36 ± 7%		
+33 0-2 A	± 33 V REG LOAD:	FL	0.75 ± 9%	33 ± 5%	+33 V Red (+)
		NL	0		+33 V Black (Ret)
		CCL	1.0 ± 7%		
-33 V 0-2 A	± 33 V REG LOAD:	FL	0.75 ± 9%	33 ± 5%	-33 V Red (-)
		NL	0		-33 V Black (Ret)
		CCL	1.0 ± 7%		
+29 V 0-20 A	+29 V,6 V REG LOAD:	FL	10.0 ± 8%	29 ± 6%	+29 V Red (+)
		NL	0		+29 V Black (Ret)
		CCL	11.6 ± 8%		
+6 V 0-5 A	+29 V,6 V REG LOAD:	FL	*2.47 ± 15%	6 ± 10%	+6 V Red (+)
		NL	0		+6 V Black (Ret)
		CCL	*2.9 ± 15%		

*Readings made on 0-5 scale. All other M1 readings are on 0-2 scale.
**Voltages indicated on M2 are available on front panel at VDC terminals (TP27 and TP28).

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3.5.1.7 OVERLOAD TEST circuitry, 0-2 A LOAD.

3.5.1.7.1 MIN load. AMPERES and VOLTS meters shall indicate 0.018 amperes ± 17% and 10 V ± 7%, respectively, under the following conditions :

a. TEST SELECTOR switch S13 in POWER SUPPLY position.

b. 10 V ± 3% between J8-B(+) and J8-C(Ret).

c. POWER SUPPLY switch S12 in +6 V 0-5 A position.

d. +29 V, 6 V REG LOAD switch S9 in NL position.

e. 2 A CB circuit breaker CB3 closed (White band not visible).

f. A shorting wire connected between 0-2 A LOAD (Black) and +6 V (Black).

g. A shorting wire connected between 0-2 A LOAD (Red) and +6 V (Red).

h. 0-2 A LOAD switch S11 in MIN position.

i. 0-2 A LOAD variable control R29 fully counterclockwise.

3.5.1.7.2 MAX load. AMPERES meter indication shall be variable from 0.19 to 2.5 amperes ± 11% by operation of 0-2 A LOAD variable control R29 and VOLTS meter shall indicate 10 V ± 7% under the following conditions:

a. Controls and conditions met in accordance with 3.5.1.7.1 a. through f.

b. A 4 ohms ± 5% load connected between 0-2 A LOAD (Red) and +6 V (Red).

c. 0-2 A LOAD switch S11 is MAX position.

The 2.5 AMPERES meter indication shall occur when R29 is fully clockwise.

3.5.1.8 2 A CB circuit breaker CB3. The 2 A CB circuit breaker (CB3) shall be closed when reset pushbutton is depressed so white band is not exposed. The circuit-breaker shall be open when reset pushbutton is pulled out to expose white band.

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3.5.1.9 OVERLOAD TEST circuitry, 0-15 A LOAD.

3.5.1.9.1 MIN load. AMPERES and VOLTS meter shall indicate 0.73 amperes ± 17% and 24 V ± 7% respectively, under the following conditions:

a. TEST SELECTOR switch S13 in POWER SUPPLY position.

b. 24 V ± 3% between J9-X(+) and J9-Y(Ret).

c. POWER SUPPLY switch S12 in +29 V 0-2 A position.

d. +29 V, 6 V REG LOAD switch S9 in NL position.

e. 15 A CB circuit breaker CB2 closed (white band not visible).

f. A shorting wire connected between 0-15 A LOAD (Black) and +29 V (Black).

g. A shorting wire connected between 0-15 A LOAD (Red) and +29 V (Red).

h. 0-15 A LOAD switch S10 in MIN position.

i. 0-15 A LOAD variable control R27 fully counterclockwise.

3.5.1.9.2 MAX load. AMPERES meter indiction shall be variable from 4.8 to 12.0 amperes ± 11% by operation of 0-15 A LOAD variable control R27 and VOLTS meter shall indicate 24 V ± 7% under the following conditions:

a. Controls and conditions met in accordance with 3.5.1.9.1a. through f.

b. A 2 ohm ± 5% load connected between 0-15 A LOAD (Red) and +29 V (Red).

c. 0-15 A LOAD switch in MAX position

The 12 ampere meter indication shall occur when R27 is fully clockwise.

3.5.1.10 15 A CB circuit breaker CB2. The 15 A CB circuit breaker (CB2) shall be closed when reset pushbutton is depressed so white band is not exposed. The circuit breaker shall be open when reset pushbutton is pulled out to expose white band.

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3.5.1.11 SERVO 115 V, 400 CPS switch S21.

3.5.1.11.1 OFF legend. The OFF legend of SERVO 115 V, 400 CPS switch S21 shall be illuminated under the following conditions:

a. 115 V ± 2 Vac, 400 ± 20 Hz applied between J10-C and J10-D (Ret).

b. S21 in normally closed (OUT) position.

c. AC circuit breaker CB4 closed (white band not visible).

3.5.1.11.2 ON legend. The ON legend of S21 shall be illuminated and 115 ± 2 Vac, 400 ± 20 HZ shall be available between J1-E/J1-F(Ret), J7-D/J1-F(Ret) and J7-s/J7-i(Ret) under the following conditions:

115 V ± 2 Vac, 400 ± 20 HZ applied between J10-C and J10-D (Ret).

b. S21 in normally open (IN) position.

c. 24 Vdc ± 5% applied between J10-A(+) and J10-I(Ret).

d. AC circuit breaker CB4 closed (white band not visible).

3.5.1.12 AC circuit breaker CB4. The AC circuit breaker CB4 shall be closed when reset pushbutton is depressed so white band is not exposed. The circuit breaker shall be open when reset pushbutton is pulled out to expose white band.

3.5.2 Servo electronics section.

3.5.2.1 Servo electronics switch circuitry.

3.5.2.1.1 SERVO COMMAND switch S5.

3.5.2.1.1.1 SERVO COMMAND legend. The SERVO COMMAND legend of S5 shall be illuminated when 24 Vdc ± 5% is applied to J7-p(+) and J10-I(Ret).

3.5.2.1.1.2 SERVO COMMAND switch. 24 Vdc ± 5% shall be available between J6-W(+) and J6-X(Ret) under conditions stated in 3.5.1.2.1 and S5 in normally open" (IN) position. With S5 in normally closed (OUT) position, 0.0 ± 0.4 Vdc shall be present between J6-W(+) and J6-X(Ret).

3.5.2.1.2 ELEVATION ON switch S22. ELEVATION ON legend of S22 shall be illuminated and 24 Vdc ± 5% shall be available between J7-V(+) and J1-p(Ret) when 24 Vdc ±5% is applied between J7-p(+) and J1-p(Ret) and S22 is in normally open (IN) position. With S22 in normally closed (OUT) position the ELEVATION ON legend of S22 shall be extinguished and 0.0 ± 0.4 Vdc shall be present between J7-V(+) and J1-p(Ret).

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3.5.2.1.3 TRAVERSE ON switch S23. TRAVERSE ON legend of S23 shall be illuminated and 24 Vdc ± 5% shall be available between J7-H(+) and J1-p(Ret) when 24 Vdc ± 5% is applied between J7-p(+) and J1-p(Ret) and S23 in normally open (IN) position. With S23 in normally closed (OUT) position, the TRAVERSE ON legend of S23 shall be extinguished and 0.0 ± 0.4 Vdc shall be present between J7-H(+) and J1-p(Ret).

3.5.2.1.4 MODE SELECTION NORMAL/STOW switch S27.

3.5.2.1.4.1 NORMAL legend. NORMAL legend of S27 shall be illuminated and an infinite resistance shall exist between points listed below under conditions as stated in 3.5.1.2.1 and S27 in normally closed (OUT) position.

J10-I and J7-G
J10-I and J1-U

3.5.2.1.4.2 STOW legend. STOW legend of S27 shall be illuminated and continuity shall exist between points listed below under conditions as stated in 3.5.1.2.1 and S27 is in normally open (IN) position.

J10-I and J7-G
J10-I and J1-U

3.5.2.1.5 MODE SELECTION UP-RIGHT/DOWN-LEFT switch S28.

3.5.2.1.5.1 UP-RIGHT legend. UP-RIGHT legend of S28 shall be illuminated and continuity shall exist between J7-d and J1-U when conditions stated in 3.5.2.1.4.2 are met and S28 is in normally closed (OUT) position.

3.5.2.1.5.2 DOWN-LEFT legend. DOWN-LEFT legend of S28 shall be illuminated and an infinite resistance shall exist between J7-d and J1-U when conditions stated in 3.5.2.1.4.2 are met and S28 is in normally open (IN) position.

3.5.2.1.6 MODE SELECTION LIMIT ON switch S29. The LIMIT ON legend OF S29 shall be illuminated and continuity shall exist between points listed below under conditions stated in 3.5.1.2.1 and S29 Is in normally open (IN) position. With S29 in normally closed (OUT) position the LIMIT ON legend of S29 shall be extinguished and an infinite resistance shall exist between points listed below:

J10-I and J1-S
J10-I and J7-J
J10-I and J7-W

3.5.2.1.7 LOAD switch S20.

3.5.2.1.7.1 LOW legend. Under conditions stated in 3.5.1.2.1 and S20 in normally closed (OUT) position, the LOW legend of S20 shall be illuminated.

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3.5.2.1.7.2 HIGH LOAD legend. Under conditions stated in 3.5.1.2.1, S20 in normally open (IN) position and a shorting wire between J1-Y and J1-A, the following conditions exist:

a. HIGH legend of S20 illuminated.

b. Resistance between J1-m and J1-r shall be 1 ohm + 2%.

3.5.2.1.8 AC SIG switch S24.

3.5.2.1.8.1 ELEVATION legend. The ELEVATION legend of S24 shall be illuminated under the following conditions:

a. Controls and conditions met in accordance with 3.5.1.11.2.

b. A shorting wire between J7-E and J7-D.

c. S24 in normally closed (OUT) position.

3.5.2.1.8.2 TRAVERSE legend. The TRAVERSE legend of S24 shall be illuminated under the following conditions:

a. Controls and conditions met in accordance with 3.5.1.11.2.

b. A shorting wire between J7-E and J7-D.

c. S24 in normally open (IN) position.

3.5.2.1.9 DC SIG switch S25.

3.5.2.1.9.1 ELEVATION legend. The ELEVATION legend of S25 shall be illuminated under the following conditions:

a. Controls and conditions met in accordance with 3.5.1.11.2.

b. A shorting wire between J7-E and J7-D.

c. S25 in normally closed (OUT) position.

3.5.2.1.9.2 TRAVERSE legend. The TRAVERSE legend of S25 shall be illuminated under the following conditions:

a. Controls and conditions met In accordance with 3.5.1.11.2.

b. A shorting wire between J7-E and J7-D.

c. S25 In normally open (IN) position.

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3.5.2.2 Signal sources (SERVO ELECTRONICS section). The voltages shall be as indicated hereafter under the following conditions:

a. Controls and conditions met in accordance with 3.5.1.2.1.

b. Controls and conditions met in accordance with 3.5.2.1.8.1.

c. Controls and conditions met in accordance with 3.5.2.1.9.1.

d. 18K ohm + 5%, resistive load between J7-X and J7-Y.

e. 4.7K ohm ± 5%, resistive load between J7-A and J7-B.

3.5.2.2.1 AC signal source.

a. AC SIGNAL rotary switch S31 in 0° position. with S31 in 0° position, the voltage across IN(TP23) and RET(TP24) terminals shall be variable by operation of AC SIGNAL potentiometer R58 from O to not less than 0.45 VRMS, 400 Hz, in phase ± 15° with respect to J10-C. O volt occurs at fully counterclockwise position of R58.

b. AC SIGNAL rotary switch S31 in 180° position. With S31 in 180° position, the voltage across IN(TP23) and RET(TP24) terminals shall be variable by operation of AC SIGNAL potentiometer R58 from O to not less than 0.45 VRMS, 400 Hz, 180° ± 15° out of phase with respect to J10-C.

c. AC SIGNAL switch S31 in SHORT position. With S31 in SHORT position and AC SIGNAL potentiometer R58 fully counterclockwise, the voltage across IN(TP23) and RET(TP24) terminals shall be less than 0.1 millivolt RMS.

3.5.2.2.2 DC Signal source.

a. DC SIGNAL rotary switch S26 in + position. With S26 in + position, the voltage across IN(TP25) and RET(TP26) terminals shall be variable by operation of DC SIGNAL potentiometer R60 from 0 to not less than +4.25 Vdc. 0 volt is present when R60 is fully counterclockwise. The 400 Hz ripple on the dc signal shall be less than 0.2 volts p-p.

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b. DC SIGNAL rotary switch S26 in - position. With S26 in - position, the voltage across IN(TP25) and RET(TP26) terminals shall be variable by operation of DC SIGNAL potentiometer R60 from 0 to not less than -4.25 Vdc. 0 volt is present when R60 is fully counterclockwise. The 400 Hz ripple on the dc signal shall be less than 0.2 volts p-p.

c. DC SIGNAL switch S26 in SHORT position. With S26 in SHORT position and DC SIGNAL potentiometer R60 fully counterclockwise, the voltage across IN(TP25) and RET(TP26) terminals shall be less than 1.0 millivolt dc.

3.5.3 Fire control servo electronics section.

3.5.3.1 280 V, 15 V switch S32. Continuity shall exist between J10-A and J11-F when S32 is in normally open (IN) position. Resistance between J10-A and J11-F shall be infinite when S32 is in normally closed (OUT) position.

3.5.3.1.1 280 V OA, 280 V OB legend. The 280 V OA and 280 V OB legends of S32 shall be illuminated under the following conditions:

a. 115 V ± 2 Vac, 400 ± 20 Hz applied between J11-B and J11-C.

b. OA circuit breaker CB5 closed (white band not visible).

c. OB circuit breaker CB6 closed (white band not visible).

3.5.3.1.2 +15 V legend. The +15 V legend of S32 shall be illuminated and 15 Vdc shall be present between J4-t(+) and J4-e(Ret) when 15 Vdc ± 3% is applied between J11-R(+) and J11-T(Ret) and +15 circuit breaker CB7 is closed (white band not visible).

3.5.3.1.3 -15 v legend, The -15 V legend of S32 shall be illuminated and 15 Vdc shall be present between J4-m(-) and J4-e(Ret) when 15 Vdc ± 3% is applied to J11-S(-) and J11-T(Ret) and -15 circuit breaker CB8 is closed (white band not visible).

3.5.3.2 280 V P-P SQ WV 400 CPS OA circuit breaker CB5. The OA circuit breaker CB5 shall be closed when reset pushbutton is depressed so white band is not exposed. The circuit breaker shall be open when reset pushbutton is pulled out to expose the white band.

3.5.3.3 280 V P-P SQ MV 400 CPS OB circuit breaker CB6. The OB circuit breaker CB6 shall be closed when reset pushbutton is depressed so white band is not exposed. The circuit breaker shall be open when reset pushbutton is pulled out to expose the white band.

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3.5.3.4 +15 circuit breaker CB7. The +15 circuit breaker CB7 shall be closed when reset pushbutton is depressed so white band is not exposed. The circuit breaker shall be open when reset push-button is pulled out to expose the white band.

3.5.3.5 -15 circuit breaker CB8. The -15 circuit breaker CB8 shall be closed when reset pushbutton is depressed so white band is not exposed. The circuit breaker shall be open when reset push-button is pulled out to expose the white band.

3.5.3.6 Signal sources (FIRE CONTROL ELECTRONICS section). The voltages across IN(TP29) and RET(TP30) terminals or J4-C and J4-e shall be as indicated hereafter under the following conditions:

a. 56 K ohm ± 5%, 1/2 watt resistive load between J4-C and J4-S.

b. +15 circuit breaker CB7 closed (white band not visible).

c. -15 circuit breaker CB 8 closed (white band not visible).

d. 15 Vdc ± 3% applied to J11-R(+) and J11-T(Ret).

e. 15 Vdc ± 3% applied to J11-S(-) and J11-T(Ret).

3.5.3.6.1 DC SIGNAL/PULSE rotary switch S30 in DC SIGNAL, +13 position. with S30 in DC SIGNAL +13 position, the voltage across IN(+) and RET(Ret) shall be variable by operation of AMPL DC INPUT control R67 from 0 to not less than 13 vdc. 0 volt is present when R67 is fully counterclockwise.

3.5.3.6.2 DC SIGNAL/PULSE rotary switch S30 in DC SIGNAL -13 position. With S30 in DC SIGNAL -13 position, the voltage across IN(+) and RET(Ret) shall be variable by operation of AMPL DC INPUT control R67 from 0 to not less than -13 Vdc. 0 volt is present when R67 is fully counterclockwise.

3.5.3.6.3 DC SIGNAL/PULSE rotary switch S30 in SHORT position. With S30 In SHORT position, 0 volts exist between IN(+) and RET(Ret) and continuity exists between J4-C and J11-T.

3.5.3.6.4 DC SIGNAL/PULSE rotary switch S30 in PULSE +13 position. with S30 in PULSE +13 position, the voltage waveform between J4-C and J4-e shall be per line 1 of Table VI.

3.5.3.6.5 DC SIGNAL/PULSE rotary switch S30 In PULSE -13 position. With S30 in PULSE -13 position, the voltage waveform between J4-C and J4-e shall be per line 2 of Table VI.

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TABLE VI. Pulse voltage characteristics

LINE	LIMITS VDC	1/2 CYCLE PERIOD SECONDS NEG	POS	RISE TIME MILLISEC	FALL TIME MILLISEC
1	0 +1.0 0 13 ± 1	0.32 to 0.65	0.32 to 0.65	less than 0.1	less than 0.1
2	0 ± 1.5 13 ± 1	0.32 to 0.65	0.32 to 0.65	less than 0.1	less than 0.1

3.6 First article. When specified in the contract or purchase order, sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.7 Soldering. All soldering shall be in accordance with the general requirements of MIL-STD-2000 in addition to detail requirement (task) F of paragraph 1.2 of MIL-STD-2000.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible fo the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspections set forth in this specification shall become part of the contractor’s overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions . The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Special sampling inspection (see 4.5).

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a.	Temperature	+60°F to +90°F (+16°C to +32°C)
b.	Relative humidity	90% RH max.
c.	Pressure	28 to 32 inches of Hg (700 to 815 mmHg)

4.3 First article Inspection.

4.3.1 Sample. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the assemblies, components and test specimens listed below in the quantities indicated.

Name	Drawing	Quantity
Power Supply and Servo Test Panel	10554530	3

4.3.2 Inspections to be performed. As determined by the Government, the first article assemblies, components and test specimens may be subjected to any or all of the examinations and tests specified in this detail specification and be inspected for compliance with any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component, or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected The Government reserves the right to terminate inspection upon any failure of any assembly, component or test specimen to comply with any of the requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the

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Inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations and tests.

4.4.2.1 Components, assemblies and subassemblies.

a. The contractor's quality program or detailed inspection plan shall provide assurance of compliance of all characteristics with the applicable drawings and specifications utilizing as a minimum the conformance criteria specified in this document and the Quality Assurance Provisions (QAP) listed in the Technical Data package (TDP). In the absence of QAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.

b. Alternative inspection procedures, methods or equipment, such as statistical process control, tool control, other types of sampling procedures, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them In a written proposal submitted to the procuring contracting officer for evaluation and approval by the Government. When required, the contractor shall demonstrate that the effectiveness of the proposed alternative(s) is equal to or better than the specified quality assurance provisions herein. In cases of dispute as to whether the contractor's proposed alternative(s) provide equal assurance, the provisions of this specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program plan or detailed inspection system, as applicable.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with the sampling plans specified in 4.4.2.3 and 4.5 herein. Examination and test for packaging and marking shall be in accordance with Section 5 herein. The tabulated classification of defects in 4.4.2.3 shall constitute the minimum inspection to be performed by the contractor after first article approval and prior to Government acceptance or rejection by item or lot.

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4.4.2.3 Functional tests. The tests in Table VII shall be performed on a 100% basis.

TABLE VII. Functional tests (100 percent inspection).

CHARACTERISTIC		REQUIREMENT	TEST PROCEDURE
Critical:		None	N/A
Major:			
101.	Electrical	3.4	4.7.2
102.	Performance	3.5	4.7.3
Minor:		None	N/A

4.5 Special sampling. One Power Supply and Test Panel out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests of Table VII and shall then meet the tests in Table VIII. Unless otherwise specified in the contract, the items shall be returned to the lot upon successful completion of testing.

TABLE VIII. Special sampling

CHARACTERISTIC		REQUIREMENT	TEST PROCEDURE
Critical:		None	N/A
Major:			
103.	Vibration	3.3.1	4.7.1.1
104.	Shock	3.3.2	4.7.1.2
105.	Altitude	3.3.3	4.7.1.3
106.	Temperature	3.3.4	4.7.1.4
107.	Humidity	3.3.5	4.7.1.5
Minor:		None	N/A

4.5.1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract.

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4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the “Methods of Inspection” specified in 4.7 of this specification and with the component inspection procedures specified in "Examination" and “Test Facilities” requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is Intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2, the test equipment listed in Table IX (Test equipment list) shall be used in the performance of the applicable test methods and procedures as specified in 4.7.

TABLE IX. Test equipment list.

ITEM NUMBER	DESCRIPTION
1. Power supply	0-28 Vdc ± 5%, 10 A, unregulated, less than 5% ripple
2. Power supply	0-40 Vdc ± 3% 17 A, unregulated, less than 5% ripple

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TABLE IX. Test equipment list (continued)

ITEM NUMBER		DESCRIPTION
3.	Volt measuring device, ac-dc	Range - ac: 0 to 120 - dc: 0 to 60 Accuracies of ± 2% on dc and ± 3% on ac Sensitivities: 20,000 ohms per volt dc and 5000 ohms per volt ac
4.	Resistive load (power)	2.9 ohms ± 10%, 250 watt
5.	Oscilloscope	Vertical passband dc to 10 MHz Input sensitivity: 1 mv per division Horizontal deflection: 0.2 usec per division to 0.5 sec per division
6.	W96 Cable Assembly	10556303
7.	Power supply, ac	115 ± 2 Vat, 1A, 400 Hz ±5% single phase
8.	Digital multi-meter ac/dc/ohms	Digits : 4.5 Minimum. Range: 0-20 mV min. 0-100 volts max. Accuracy: ± 0.1 percent Input Impedance: greater than 1 Megohm
9.	Power supply, dc	+15 Vdc ± 3%, 1.3 A, regulated, less than 1% ripple
10.	Power supply, dc	-15 Vdc ± 3%, 1.3 A, regulated, less than 1% ripple
11.	Resistive load	4 ohms ± 5%, 50 Watts
12.	Resistive load (1/2W, ± 5%)	(a) 18 K ohm (b) 4.7 K ohm (c) 56 K ohm
13.	Resistive load	2 ohms ± 5%, 500 Watt
14.	Power supply	23 Vdc ± 3%, 2 A, unregulated
15.	Power supply	23 Vdc ± 3%, 2 A, unregulated
16.	Power supply	33 Vdc ± 3%, 1.5 A, unregulated
17.	Power supply	33 Vdc ± 3%, 1.5 A, unregulated
18.	Power supply	6 Vdc ± 5%, 3.0 A, unregulated

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4.7 Methods of inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance, and individual test methods of MIL-STD-810 except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

4.7.1.1 Vibration. The test shall be performed as specified in Method 514, Category 8, Procedure I, of MIL-STD-810 as modified by 3.3.1 herein. In Step 1 of procedure I only steps 4 through 8 of paragraph II-2.1 apply. The Power Supply and Test Panel shall be attached to the vibration generator directly or with a fixture and securely held by its normal means of attachment. After vibration, the Power Supply and Test Panel shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1.

4.7.1.2 Shock. This test shall be performed as specified in Procedure I, Section II of Method 516 of MIL-STD-810 except that:

a. The post-test functional check (Step 5 of Procedure I) need not be conducted until completion of all shock impulses.

b. The shock pulses shall be 15 g's ± 10% amplitude and 11 ms ± 10% duration.

c. The maximum g shall occur at 5.5 milliseconds ± 10%.

d. The shock measuring system shall have a flat frequency response within ± 10% from 20 to 2,000 hertz. After the application of all the shock impulses, the Power Supply and Test Panel shall be tested as specified in 4.7.2 and 4.7.3 to determine compliance with 3.3.2.

4.7.1.3 Altitude (operating and non-operating) This test shall be performed as specified in Procedures II and III, Method 500 of MIL-STD-810. The operational checkout shall be as specified In 4.7.2 and 4.7.3 for compliance with 3.3.3.

4.7.1.4 Temperature.

4.7.1.4.1 High (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 501 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.4.2 Low (operating and non-operating) . This test shall be performed as specified in Procedures I and II, Method 502 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

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4.7.1.5 Humidity (non-operating) This test shall be performed in accordance with procedure III, Method 507 of MIL-STD-810 except that in Step 7, checkout shall be performed at the conclusion of the test only, with the item at standard ambient conditions. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.5.

4.7.2 Electrical.

4.7.2.1 Continuity. Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3*4.1*

4.7.2.2 Resistance. Resistance shall be measured with an ohmmeter (Table IX, Item 8) for compliance with 3.4.2 and as given herein.

4.7.2.2.1 OVERLOAD TEST circuitry. Verify the resistance values at the terminal points and with the control settings of Table I using ohmmeter (Table IX, item 8).

4.7.2.2.2 Converter loading circuitry. Set TEST SELECTOR switch S13 to CONVERTER position. Use ohmmeter (Table IX, item 8) to verify the resistance values between Vdc terminals (TP27 and TP28) of-the power supply section with control settings and results as given in Table II.

4.7.2.2.3 Servo loading circuitry. Verify the resistance values at the terminal points and control settings of Table III using ohmmeter (Table IX, item 8).

4.7.3 Performance.

4.7.3.1 Power supply test section.

4.7.3.1.1 +24 VDC switch S1.

4.7.3.1.1.1 OFF legend. Apply 24.0 ± 1.2 Vdc using power supply (Table IX, item 1) between J10-A(+) and J10-I(Ret). Set S1 to normally closed (OUT) position. verify that OFF legend is illuminated.

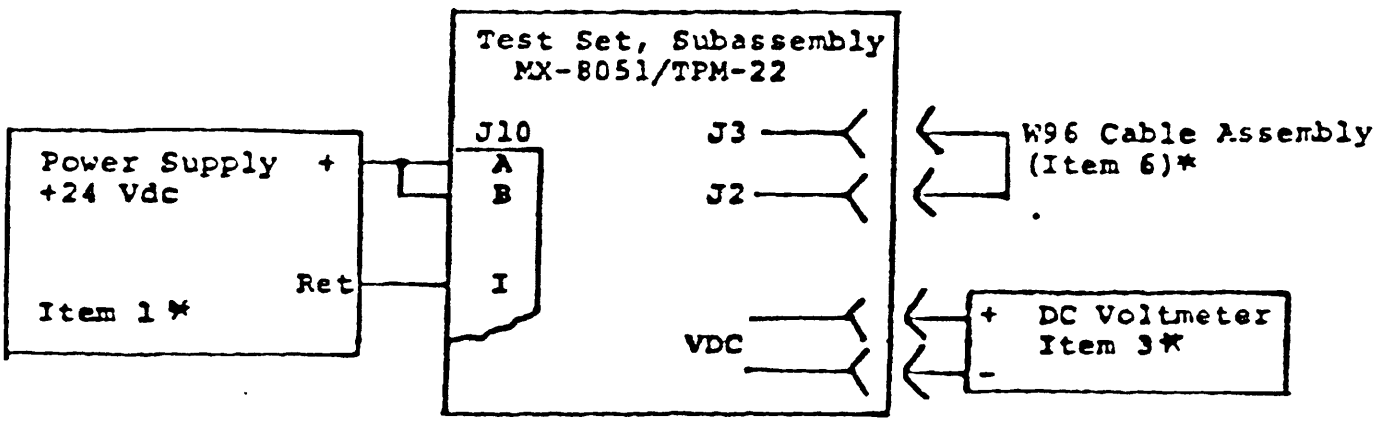
4.7.3.1.1.2 ON legend. under conditions as stated in 4.7.3.1.1.1, operate S1 to the normally open (IN) position and verify that the ON legend is illuminated.

4.7.3.1.2 TEST MODE switch S4.

4.7.3.1.2.1 POWER SUPPLY CONV & SERVO legend. Using power supply (Table IX, item 1), set up conditions as stated in 3.5.1.2.1a. through c. Set S4 to the normally closed (OUT) position and verify that POWER SUPPLY CONV & SERVO legend Is illuminated. Using voltmeter (Table IX, item 3), verify that 24.0 ± 1.2 Vdc is present between J5-A(+) and J5-B(Ret).

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4.7.3.1.2.2 REGULATOR legend. Use test setup as shown in Figure 1. Set S4 to the normally open (IN) position and verify the REGULATOR legend is illuminated. Verify voltages specified in Table IV are present at VOLTS meter (M2) for each switch setting specified in Table IV.



*Refer to 4.6.2.3

FIGURE 1. Power supply and servo test panel test setup.

4.7.3.1.3 DC circuit breaker CB1. Close DC circuit breaker (CB1) by depressing reset pushbutton so white band is not exposed. Connect dc power using power supply (Table IX, item 1) and set up conditions as stated in 3.5.1.2.1a. through d. Observe that POWER SUPPLY CONV & SERVO legend of TEST MODE switch is illuminated. Open circuit breaker CB1 by pulling out reset pushbutton so that white band is exposed. observe that POWER SUPPLY CONV & SERVO legend of TEST MODE switch is extinguished. Turn off dc power, reset circuit breaker, and turn on dc power. Observe that POWER SUPPLY CONV & SERVO legend is again illuminated.

4.7.3.1.4 INTERLOCK switch S2.

4.7.3.1.4.1 STANDBY legend. Apply 24 Vdc ± 5%, using power supply (Table IX, item 1), between J9-G(+) and J9-H(Ret). Verify that STANDBY legend of S2 is illuminated.

4.7.3.1.4.2 OPERATE legend. Apply 24.0 ± 1.2 Vdc, using power supply (Table IX, item 1), between J9-V(+) and J9-a(Ret). Verify that OPERATE legend of S2 is illuminated. Set S2 to normally closed (OUT) position. Using DVM (Table IX, item 8), verify resistance between J9-H and J9-F is greater than 10 megohms. Set S2 to normally open (IN) position and verify resistance between J9-H and J9-F is less than 0.5 ohms.

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4.703.1.5 RADIATE COMMAND switch S3.

4.7.3.1.5.1 RADIATE COMMAND legend. Apply 24.0 ± 1.2 Vdc, using power supply (Table IX, item 1), between J8-N(+) and J8-P(Ret). Verify that RADIATE COMMAND legend of S3 is illuminated.

4.7.3.1.5.2 RADIATE COMMAND switch. Apply voltages, using power supply (Table IX, item 1), and set up conditions as stated in 3.5.1.2.1a. through d. Set S3 to normally open (IN) position. Using voltmeter (Table IX, item 3), verify that 24.0 ± 1.2 Vdc is present between J6-U(+) and J6-V(Ret). Set S3 to normally closed (OUT) position and verify 0.0 ± 0.4 Vdc is present between J6-U(+) and J6-V(Ret).

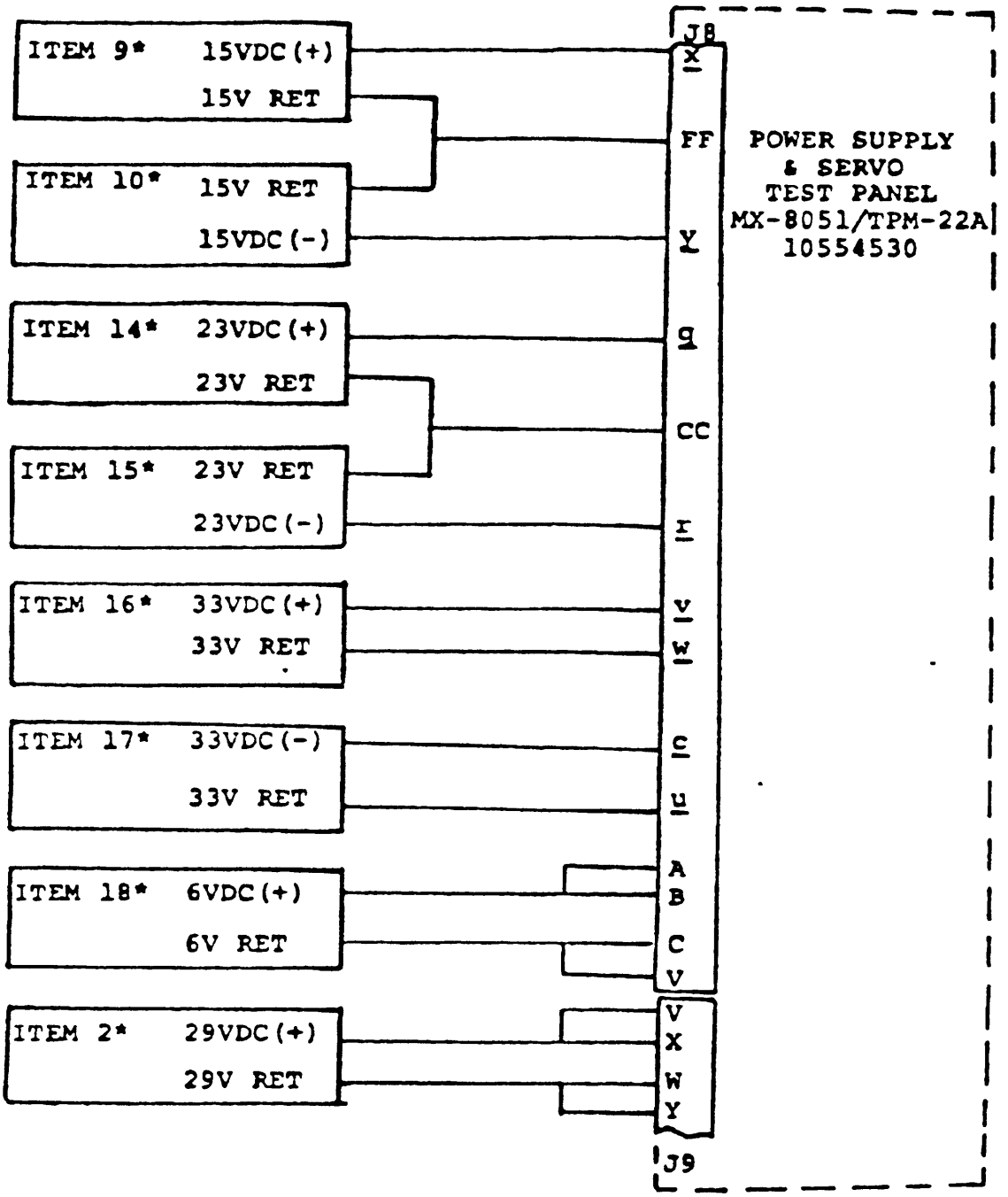
4.7.3.1.6 Regulator loading and monitoring. Apply voltages to J8 and J9 using items 2, 9, 10, 14, 15, 16, 17 and 18 as indicated in Figure 2. Connect a resistive load (Table IX, item 4) between J5-A and J5-B. Set TEST SELECTOR switch S13 to POWER SUPPLY position. Apply voltages to J10, using power supply (Table IX, item 1), and set Up conditions as stated in 3.5.1.2.1. Verify voltage readings at terminal points using voltmeter (Table IX, item 3) and meter readings as given in Table V.

4.7.3.1.7 OVERLOAD TEST circuitry, 0-2 A LOAD.

4.7.3.1.7.1 MIN load. Apply 10.0 ± 0.3 Vdc, using power supply (Table IX, item 2), between J8-B(+) and J8-C(Ret). Set conditions as stated in 3.5.1.7 and verify that AMPERES and VOLTS meters Indicate 0.018 amperes ± 17% and 10V ± 7%, respectively.

4.7.3.1.7.2 MAX load. Apply 10.0 ± 0.3 Vdc, using power supply (Table IX, item 2), between J8-B(+) and J8-C(Ret). Connect a 4 ohm resistive load (Table IX, item 11) between 0-2 A LOAD (red) and +6 V (red) terminals. Set up conditions as stated in 3.5.1.7.2 and verify that AMPERES meter indication varies continuously from 0.19 to 2.5 A + 11% when 0-2 A LOAD variable control R29 is varied from fully counterclockwise to fully clockwise. Verify that VOLTS meter Indication is 10 V ± 7%.

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* REFER TO 4.6.2.3

FIGURE 2. Regulator loading and monitoring test setup.

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4.7.3.1.8 2 A CB circuit breaker CB3. Energize 2 A CB circuit breaker CB3 by depressing reset pushbutton so white band is not exposed. Set-up conditions stated in 4.7.3.1.7.2. observe that AMPERES and VOLTS meters indicate 2.5 amperes + 11% and 10 V ± 7%, respectively. Open circuit breaker CB3 by pulling out reset push-button so that white band is exposed and observe that AMPERES meter indicates 0. Turn off dc power supply, reset circuit breaker and turn on dc power. observe that AMPERES and VOLTS indicate 2.5 amperes ± 11% and 10 V ± 7%, respectively.

4.7.3.1.9 OVERLOAD TEST circuitry, 0-15 A LOAD.

4.7.3.1.9.1 MIN load. Apply 24.0 ± 0.7 Vdc, using power supply (Table IX, item 2), between J9-X(+) and J9-Y(Ret). Set up conditions stated in 3.5.1.9.1 and verify that AMPERES and VOLTS meters indicate 0.73 amperes ± 17% and 24 V ± 7%, respectively.

4.7.3.1.9.2 MAX load. Apply 24.0 ± 0.7 Vdc, using power supply (Table IX, item 2), between J9-X (+) and J9-Y(Ret). Connect a 2 ohm resistive load (Table IX, item 13) between 0-15 A LOAD (red) and +29 V (red) terminals. Set up conditions stated in 3.5.1.9.2 and verify that AMPERES meter indication varies continuously from 4.8 to 12 amperes ± 11% when 0-15 A LOAD variable control R27 is varied from fully counterclockwise to fully clockwise. Verify that VOLTS meter indication is 24 V ± 7%.

4.7.3.1.10 15 A CB circuit breaker CB2. Energize 15 A CB circuit breaker CB2 by depressing reset pushbutton so white band is not exposed. Set up conditions stated in 4.7.3.1.9.2. Observe that AMPERES and VOLTS meters indicate 12 amperes ± 11% and 24 V ± 7%, respectively. Open circuit breaker CB3 by pulling out reset push-button so white band is exposed and observe that AMPERES meter indicates 0. Turn off dc power supply, reset circuit breaker and turn on dc power. observe that AMPERES and VOLTS meters Indicate 12 amperes ± 11% and 24 V ± 7%, respectively.

4.7.3.1.11 SERVO 115 V, 400 CPS, switch S21.

4.7.3.1.11.1 OFF legend. Apply 115 ± 2 Vac, 400 ± 20 Hz to J10 using power supply (Table IX, item 7) and set up conditions stated In 3.5.1.11.1. Verify that OFF legend of S21 is illuminated.

4.7.3.1.11.2 ON legend. Apply 115 ± 2 Vat, 400 ± 20 Hz using power supply (Table IX, item 7) and 24.0 ± 1.2 Vdc using power supply (Table IX, item 1) to J10 and set up conditions stated in 3.5.1.11.2. Verify that ON legend of S21 is illuminated. Verify, using voltmeter (Table IX, item 3) , that 115 Vac ± 5% is present between the following points:

J1-E and J1-F(Ret).

J7-D and J1-F(Ret).

J7-s and J7-i(Ret).

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4.7.3.1.12 AC circuit breaker CB4. Energize AC circuit breaker CB4 by depressing reset pushbutton so white band is not exposed. Set up conditions stated in 4.7.3.1.11.2. Observe that ON legend of SERVO 115 v 400 CPS switch S21 is illuminated. Open circuit breaker CB 4 by pulling out reset pushbutton so white band is exposed and observe that ON legend of SERVO 115V 400 CPS switch S21 is extinguished. Turn off dc and ac power supplies, reset circuit breaker and turn on dc and ac power supplies. Observe that ON legend of S21 is illuminated.

4.7.3.2 Servo electronics test section.

4.7.3.2.1 Servo electronics switch circuitry.

4.7.3.2.1.1 SERVO COMMAND switch S5.

4.7.3.2.1.1.1 SERVO COMMAND legend. Apply 24 Vdc ± 5%, using power supply (Table IX, item 1), between J7- (+) and J10-I(Ret). Verify that SERVO COMMAND legend of S5 is illuminated.

4.7.3.2.1 .1.2 SERVO COMMAND switch. Apply 24 Vdc ± 5%, using power supply (Table IX, item 1), to J10 and set up conditions stated in 3.5.1.2.1 and 3.5.2.1.1.2. Verify that 24 Vdc ± 5% is present between J6-W(+) and J6-X(Ret) using power supply (Table IX, item 3). Set S5 to normally closed (OUT) position, verify 0.0 ± 0.4 Vdc is present between J6-W(+) and J6-X(Ret).

4.7.3.2.1.2 ELEVATION ON switch S22. Apply 24 Vdc ± 5%, using power supply (Table IX, item 1), between J7- (+) and J1-p (Ret). Set S22 to normally open (IN) position. Verify that ELEVATION ON legend of S22 is illuminated. Using voltmeter (Table IX, Item 3), verify that 24.0 ± 1.2 Vdc is present between J7-V(+) and J1-p(Ret). Set S22 to normally closed (OUT) position. Verify that ELEVATION ON legend of S22 is extinguished and 0.0 ± 0.4 Vdc is present between J7-V(+) and J1-p(Ret).

4.7.3.2.1.3 TRAVERSE ON switch S23. Apply 24 Vdc ± 5%, using power supply (Table IX, item 1), between J7-p(+) and J1-p(Ret). Set S23 to normally open (IN) position. Verify that TRAVERSE ON legend of S23 is illuminated. Using voltmeter (Table IX, Item 3), verify that 24.0 ± 1.2 Vdc is present between J7-H(+) and J1-p(Ret). Set S23 to normally closed (OUT) position. verify that TRAVERSE ON legend of S23 is extinguished and 0.0 ± 0.4 Vdc is present between J7-H(+) and J1-p(Ret).

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4.7.3.2.1.4 MODE SELECTION NORMAL/STOW switch S27.

4.7.3.2.1.4.1 NORMAL legend. Apply 24.0 ± 1.2 Vdc, using power supply (Table IX, item 1), to J10 and set up conditions stated in 3.5.1.2.1. Set S27 in normally closed (OUT) position. Verify that NORMAL legend of S27 is illuminated. Using ohmmeter (Table IX, Item 8), verify that the resistance between the-following points is greater than 10 megohms:

J10-I and J7-G
J10-I and J1-U

4.7.3.2.1.4.2 STOW legend. Apply 24 .0 ± 1.2 Vdc, using power supply (Table IX, item 1), to J10 and set up conditions stated in 3.5.1.2.1. Set S27 in normally open (IN) position. Verify that STOW legend of S27 Is illuminated. Using ohmmeter (Table IX, item 8), verify that the resistance between the following points is less than 0.5 ohm.

J10-I and J7-G
J10-I and J7-U

4.7.3.2.1.5 MODE SELECTION UP-RIGHT/DOWN-LEFT switch 28.

4.7.3.2.1.5.1 UP-RIGHT legend. Apply 24.0 ± 1.2 Vdc, using power supply (Table IX, item 1) , to J10 and set up conditions stated in 3.5.2.1.4.2. Set S28 in normally closed (OUT) position. verify that UP-RIGHT legend of S28 is illuminated. Using ohmmeter (Table IX, item 8), verify that the resistance between J7-d and J1-U is less than 0.5 ohm.

4.7.3.2.1.5.2 DOWN-LEFT legend. Apply 24.0 ± 1.2 Vdc, using power supply (Table IX, item 1), to J10 and set up conditions stated In 3.5.2.1.4.2. Set S28 in normally open (IN) position. Verify that DOWN-LEFT legend of S28 is illuminated. Using ohmmeter (Table IX, Item 8), verify that the resistance between J7-d and J1-U is greater than 10 megohms.

4.7.3.2.1.6 MODE SELECTION LIMIT ON switch S29. Apply 24 Vdc ± 5%, using power supply (Table IX, item 1), to J10 and set up conditions stated in 3.5.1.2.1. Set S29 to normally open (IN) position. Verify that LIMIT ON legend of S29 is illuminated. Using ohmmeter (Table IX, item 8), verify the resistance between the following points is less than 0.5 ohm. Set S29 to normally closed (OUT) position. Verify that LIMIT ON legend of S29 is extinguished and the resistance between the points listed below is greater than 10 ohms.

J10-I and J1-S
J10-I and J7-J
J10-I and J7-W

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4.7.3.2.1,7 LOAD switch S20.

4.7.3.2.1.7.1 LOW legend. Apply 24.0 ± 1.2 Vdc, using power supply (Table IX, item 1), to J10 and set up conditions stated in 3.5.1.2.1. Set S20 in normally closed (OUT) position. verify that LOW legend of S20 is illuminated.

4.7.3.2.1.7.2 HIGH legend. Apply 24.0 ± 1.2 Vdc, using power supply (Table IX, item 1), to J10 and set up conditions stated in 3.5.1.2.1. Connect a shorting wire between J1-Y and J1-A. Set S20 in normally open (IN) position. Verify that HIGH legend of S20 is illuminated . Using ohmmeter (Table IX, Item 8), verify resistance between J1-m and J1-r is 1.00 ± 0.02 ohm.

4.7.3.2.1.8 AC SIG switch S24. Apply 24 Vdc ± 5%, using power supply (Table IX, item 1), and 115 V ± 2 V, 400 ± 20 Hz, using power supply (Table IX, item 7), to J10 and set up conditions stated in 3.5.1.11.2a. through d. Connect a shorting wire between J7-D and J7-E. Set S24 in normally closed (OUT) position. verify that ELEVATION ON legend of S24 is illuminated. Set S24 In normally open (IN) position. Verify that TRAVERSE legend of S24 is illuminated.

4.7.3.2.1.9 DC SIG switch S25. Apply 24 Vdc ± 5%, using power supply (Table IX, item 1), and 115 ± 2 Vat, 400 + 20 Hz, using power supply (Table IX, item 7), to J10 and set up conditions stated in 3.5.1.11.2a. through d. Set S25 in normally closed (OUT) position. Verify that ELEVATION legend of S25 is illuminated. Set S25 in normally open (IN) position. Verify that TRAVERSE legend of S25 is Illuminated.

4.7.3.2.2 Signal sources (SERVO ELECTRONICS section). Apply 24 Vdc ± 5%, using power supply (Table IX, item 1) and 115 ± 2 Vat, 400 ± 20 Hz, using power supply (Table IX, Item 7), to J10 as stated in 3.5.1.2.1 and 3.5.1.11.2, respectively. Connect a shorting wire between J7-D and J7-E. Connect a resistive load (Table IX, item 12a) between J7-X and J7-Y. Connect a resistive load (Table IX, item 12b) between J7-A and J7-B. Set SERVO 115 V, 400 CPS, switch S21 to illuminate ON legend. Set AC SIG switch S24 and DC SIG switch S25 to illuminate ELEVATION legends.

4.7.3.2.2.1 AC signal source.

a. AC SIGNAL rotary switch S31 In 0° position. Set S31 to 0° position. Rotate AC SIGNAL Potentiometer R58 from fully counterclockwise to fully clockwise. verify, using oscilloscope (Table IX, item 5), that signal between IN(TP23) and RET(TP24) terminals varies from O to not less than 1.26 volts peak-to-peak 400 Hz, in phase ± 15° with respect to J10-C.

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b. AC SIGNAL rotary switch S31 in 180° position. Set S31 to 180° position. Rotate AC SIGNAL potentiometer R58 from fully counterclockwise to fully clockwise. Verify, using oscilloscope (Table IX, item 5), that signal between IN(TP23) and RET(TP24) terminals varies from 0 volt to not less than 1.26 volts peak-to-peak, 400 Hz, 180° out of phase ± 15° with respect to J10-C.

c. AC SIGNAL rotary switch S31 In SHORT position. Set S31 to SHORT position. Set AC SIGNAL potentiometer R58 to fully counterclockwise position. Verify, using oscilloscope (Table IX, item 5), that voltage between IN(TP23) and RET(TP24) terminals is less than 0.33 mV peak-to-peak.

4.7.3.2.2.2 DC signal source.

a. DC SIGNAL rotary switch S26 in + position. Set S26 to + position. Rotate DC SIGNAL potentiometer R60 from fully counter-clockwise to fully clockwise. Verify, using voltmeter (Table IX, item 8), that voltage between IN(TP25) and RET(TP26) terminals varies from O volt to not less than +4.25 Vdc. Verify, using oscilloscope (Table IX, item 5), that 400 Hz ripple content on the dc signal is less than 0.2 volt peak-to-peak.

b. DC SIGNAL rotary switch S26 in - position. Set S26 to - position. Rotate DC SIGNAL potentiometer R60 from fully counter-clockwise to fully clockwise. Verify, using voltmeter (Table IX, Item 8), that voltage between IN(TP25) and RET(TP26) terminals varies from O volt to not less than -4.25 Vdc. Verify, using oscilloscope (Table IX, item 5), that 400 Hz ripple content on the dc signal Is less than 0.2 volt peak-to-peak.

c. DC SIGNAL rotary switch S26 in SHORT position. Set S26 to SHORT position and DC SIGNAL potentiometer R60 to fully counter-clockwise. Verify, using voltmeter (Table IX, item 8), that voltage across IN(TP25) and RET(TP26) terminals is less than 1.0 millivolt dc.

4.7.3.3 FIRE CONTROL SERVO ELECTRONICS section.

4.7.3.3.1 280 V, OA/280 V OB/+15 V/-l5 V switch S32. Set S32 in normally open (IN) position. Using ohmmeter (Table IX, item 8), verify that the resistance between J10-A and J11-F is less than 0.5 ohm. Set S32 in normally closed (OUT) position. Verify that the resistance between J10-A and J11-F is greater than 10 megohms.

4.7.3.3.1.1 280 V OA, 280 V OB legend. Set OA and OB circuit breakers, CB5 and CB6 to closed position (white band not visible). Apply 115 ± 2 Vat, 400 ± 20 Hz, using power supply (Table IX, item 7), between J11-B and J11-C. Verify that 280 V OA and 280 V OB legends of S32 are illuminated.

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4.7.3.3.1.2 +15 V legend. Set +15 circuit breaker C87 to closed position (white band not visible). Apply 15 Vdc ± 3%, using power supply (Table IX, item 9), between J11-R(+) and J11-T(Ret). Verify that +15 V legend of S32 is illuminated. Verify, using voltmeter (Table IX, item 3), that 15 Vdc ± 3% Is present between J4-t(+) and J4-e(Ret).

4.7.3.3.1.3 -15 V legend. Set -15 circuit breaker CB8 to closed position (white band not visible). Apply 15 Vdc ± 3% using power supply (Table IX, item 10) between J11-S(-) and J11-T(Ret). Verify that -15 V legend of S32 is illuminated. Verify, using voltmeter (Table IX, item 3), that 15 Vdc ± 3% is present between J4-m(-) and J4-e(Ret).

4.7.3.3.2 280 V P-P SQ WV 400 CPS OA and OB circuit breaker CBS and CB6. Close OA circuit breaker CBS and OB circuit breaker CB6 by depressing reset pushbutton so white band is not exposed. Apply 115 + 2 Vat, 400 ± 20 Hz, using power supply (Table IX, item 7), and setup conditions stated in 3.5.3.1.1. Observe that 280 V OA and OB legend of S32 is illuminated. Open circuit breaker CB5 by pulling out reset pushbutton so white band is exposed and observe that 280 V OA and OB legend of S32 is extinguished. Turn off ac power, reset circuit breaker and turn on ac power. 280 V OA and OB legend of S32 shall again be illuminated. Open circuit breaker CB6 by pulling out reset pushbutton so white band is exposed and observe that 280 V OA and OB legend of S32 is extinguished. Turn off ac power, reset circuit breaker and turn on ac power. 280 V OA and OB legend of S3: shall again be illuminated.

4.7.3.3.3 +15 circuit breaker CB7. Close circuit breaker CB7 by depressing reset pushbutton so white band is not exposed. Apply 15 Vdc ± 3% using power supply (Table IX, item 9) as stated in 4.7.3.3.1.2. observe that +15 V legend of S32 is illuminated. Open circuit breaker CB7 by pulling out reset pushbutton so white band Is exposed and observe that +15 V legend is extinguished. Turn off dc power, reset circuit breaker and turn on dc power. +15 V legend of S32 shall again be illuminated.

4.7. 3.3.4 -15 circuit breaker CB8. Close circuit breaker CB8 by depressing reset pushbutton so white band is not exposed. Apply 15 Vdc ± 3% using power supply (Table IX, item 10), as stated in 4.7.3.3.1.3. Observe that -15 V legend of S32 is illuminated. Open circuit breaker CB8 by pulling out reset pushbutton so white band is exposed and observe that -15 V legend is extinguished. Turn off dc power, reset circuit breaker and turn on dc power. -15 V legend of S32 shall again be illuminated.

4.7.3.3.5 Signal sources (FIRE CONTROL ELECTRONICS section). Apply 15 vdc ± 3%, using power supply (Table IX, item 9), to J11-R(+) and J11-T(Ret) and 15 Vdc ± 3%, using power supply (Table IX, item 10), to J1-S(-) and J11-T(Ret). Connect a resistive load (Table IX, item 12c) between J4-C and J4-s. Close +15 and -15 circuit breakers CB7 and CB8.

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4.7.3.3.5.1 DC SIGNAL/PULSE rotary switch S30 in DC SIGNAL, +13 position. Rotate AMPL DC INPUT control R67 from fully counter-clockwise to clockwise position. Verify, using DVM (Table IX, item 8), that voltage between IN(+) and RET(Ret) varies from 0 to not less than 13 Vdc.

4.7.3.3.5.2 DC SIGNAL/PULSE rotary switch S30 in DC SIGNAL, -13 position. Rotate AMPL DC INPUT control from fully counterclockwise to clockwise position. Verify, using DVM (Table IX, item 8), that voltage between IN(-) and RET(Ret) varies from 0 to not less than 13 Vdc.

4.7.3.3.5.3 DC SIGNAL/PULSE rotary switch S30 in SHORT position. Verify, using DVM (Table IX, item 8), that 0 Vdc exists between IN(+) and RET(Ret) and also verify, using ohmmeter (Table IX, item 8), that resistance between J11-T and J4-C is less than 0.5 ohm.

4.7.3.3.5.4 DC SIGNAL/PULSE rotary switch S30 in PULSE, +13 position. Verify, using oscilloscope (Table IX, item 5), that signal between J4-C(+) and J4-e(Ret) conforms to data on line 1, Table VI.

4.7.3.3.5.5 DC SIGNAL/PULSE rotary switch S30 in PULSE, -13 position. Verify, using oscilloscope (Table IX, item 5), that signal between J4-C(+) and J4-e(Ret) conforms to data on line 2 of Table VI.

4.8 Soldering. Soldering processes shall be inspected by certified inspectors to the requirements and acceptance criteria of MIL-STD-2000 as specified in 3.7.

5. PACKAGING

5.1 General. Preservation, packing, and marking shall be in accordance with Packaging Data Sheet PS 10554530. The level of protection shall be as specified in the procurement document.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The Power Supply and Servo Test Panel is a part of Radar Test Set AN/TPM-22A. It is used to check the performance of and assist in isolating malfunctions in the following power supplies and servo electronics used in Radar Set AN/VPS-2.

a. Power supply subassembly (5A3) for unit 5 and its subunits (converter and regulators).

b. System control functions of unit 5.

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c. Elevation and traverse servo amplifiers 5A2 and 5A1 of unit 5, power supply PP-4812/VPS-2.

d. Torque current and magnet current amplifiers of the fire control servo system.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Applicable stock number.

c. First Article Sample requirements.

d. Packaging requirements, if other than specified in Section 5.

e. Serialization requirements, if applicable.

f. Certificate of Conformance for each lot or shipment of product.

g Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

6.3 Submission of contractor inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army AMCCOM, ATTN: AMSMC-QAF-I , Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC), may-also include drawings prepared by, and identified as, U.S. Army Armament, Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.5 Subject term (key word) listing.

Radar Test Set AN/TPM-22A
Servo amplifiers

6.6 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
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Preparing activity:
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INSTRUCTIONS

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1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
MIL-T-50335B (AR)	30 JANUARY 1992

3. DOCUMENT TITLE: TEST SET, POWER SUPPLY: MX-8051-TPM-22A

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
US ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial (201) 724-6674	(2) AUTOVON 880-6674
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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